WEEK 13 • IS321 ENTERPRISE ARCHITECTURE

The EA³ Cube Framework

Revisiting Bernard's Model in Full — Components, Threads, and Segments

TODAY'S ROADMAP

What We'll Cover

01	**Why Revisit the Cube?**	You've used it three times already — now we go deeper
02	**Vertical & Horizontal Components**	What actually lives inside each level, what cuts across all of them, and how Levels, Segments & Artifacts combine
03	**Current vs. Future Architecture**	The Architecture Management Plan as the bridge between the two
04	**Threads, Segments & Comparisons**	Security/Standards/Skills, how Lines of Business fit into the cube, and how Segments differ across enterprise types
05	**Worked Examples & Recap**	Amanah Bank and the university online-degree example, the Cube-to-Repository crosswalk, and common pitfalls

FOUNDATIONS

Why Revisit the Cube?

You've already met the EA³ Cube three times this semester — each time from a different angle:



Week 3

Introduced the five levels themselves — Goals & Initiatives through Networks & Infrastructure.



Week 5

Used the same levels (plus Security Solutions) as the EA Repository's six columns.



Week 11

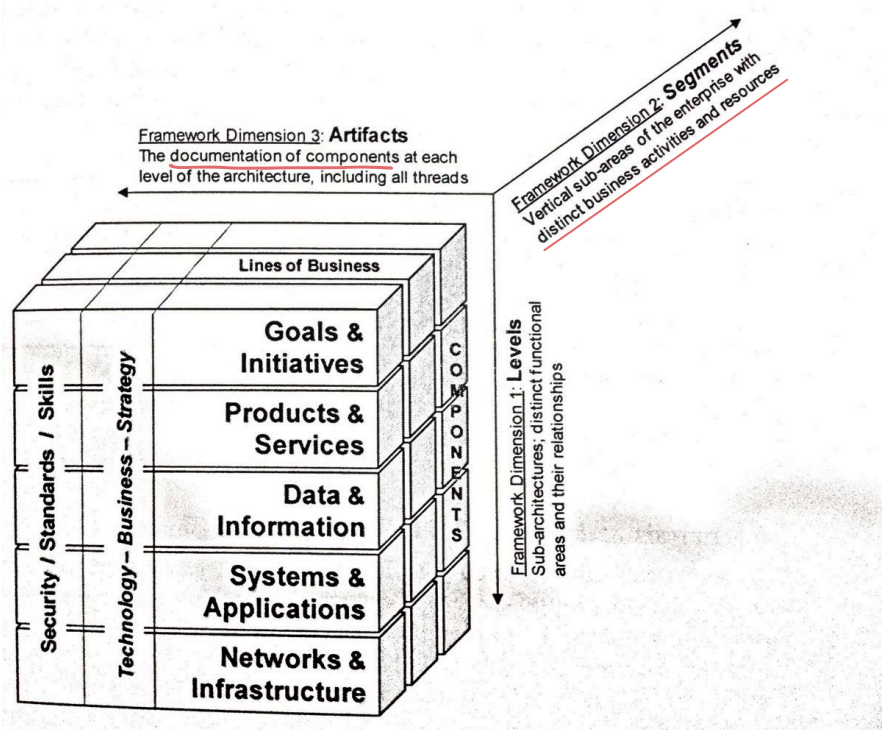
Positioned the Cube alongside Zachman, TOGAF, and DoDAF in the framework comparison.



Today

The full picture: vertical/horizontal components, current vs. future architecture, threads, and segments — the depth we skipped until now.

The EA³ Cube: Three Dimensions



Three dimensions at once

Levels

The five hierarchical layers (Week 3).

Segments

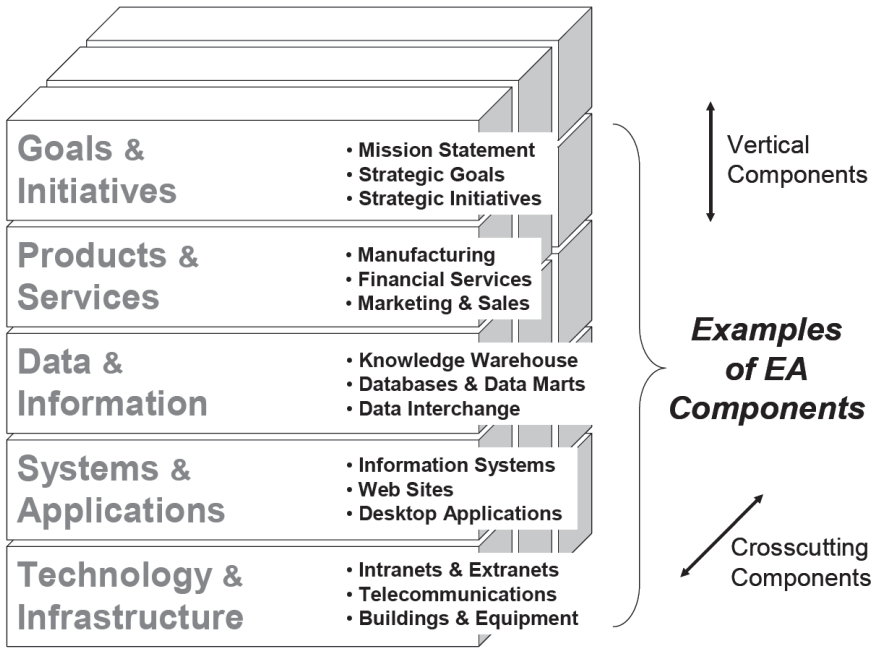
Vertical sub-areas — Lines of Business. → sub-architecture slice

Artifacts

The documentation at each level.

NEW DEPTH — NOT COVERED BEFORE

Vertical & Horizontal Components



Two kinds of components

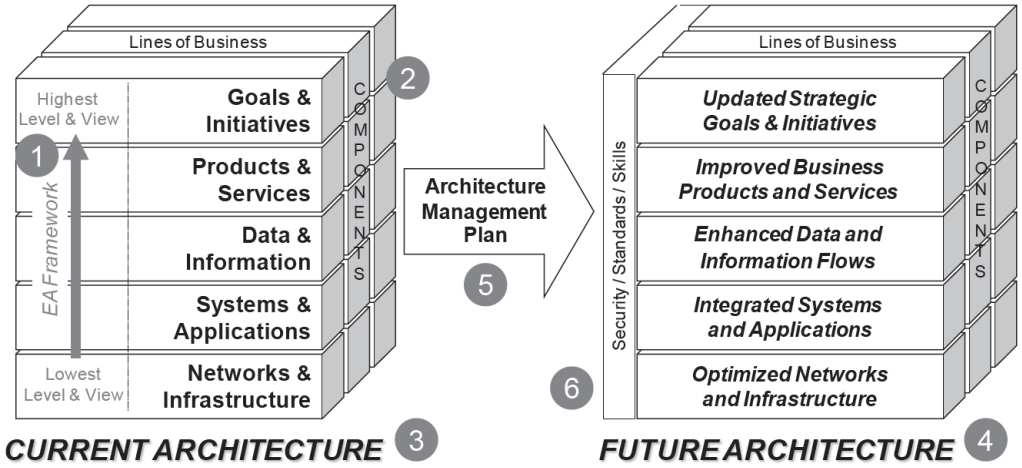
Vertical: examples specific to one level (Mission Statement at Goals & Initiatives; Databases at Data & Information).

serves multiple LoBs

Crosscutting: examples that span every level at once — Intranets, telecommunications, buildings & equipment.

NEW DEPTH — TIES BACK TO WEEK 2

Current vs. Future Architecture



1 EA framework
2 components per level
3 current Architecture
4 Architecture Mgmt
5 Future Arch
6 3 threads

This figure is literally Week 2's six elements, seen from the Cube's angle: (1) the EA Framework spans lowest to highest view, (2) Components at each level, (3) Current Architecture, (4) the Architecture Management Plan bridges to (5) Future Architecture, with (6) Security/Standards/Skills threaded through both sides.

CROSS-CUTTING

Threads: Security, Standards & Skills

Threads run through every level of the Cube at once — not their own level, but a concern that touches all five. Already glimpsed in Week 5 (the Security Solutions repository column); here's the full picture.

Security

Protecting data, systems, and infrastructure at every level — from strategic risk down to network firewalls.

Standards

The technical and business standards each level's components must comply with — consistency across the whole enterprise.

Skills

The workforce capabilities needed to build, operate, and govern the architecture at every level.

THE SECOND DIMENSION

Segments: Lines of Business

↓ each has its own five-level documentation

Segments are the Cube's second dimension — vertical sub-areas of the enterprise with distinct business activities and resources, shown as "Lines of Business" slabs stacked at the top of the cube figure. Each Line of Business gets its own complete five-level documentation.

CONNECTS TO

- Week 7's Organizational Network Model — Functional Teams map directly onto Segments/Lines of Business.
- Week 7's segmented approach to documenting large enterprises — vertical/horizontal/extended components.
- Week 5's EA Repository — each Segment can maintain its own repository following the same six-column structure.

→ documenting VC is more cost-effective

WORKED EXAMPLE

Amanah Bank: One Cell of the Cube

Back to Amanah Bank's open-banking initiative (Weeks 3, 4, 9) — this time located precisely within all three Cube dimensions at once.

Level

Products & Services (Level 2) — the new API-based account access product.

Segment

Retail Banking — one Line of Business among several at Amanah Bank.

Artifact

The Business Service/Function Catalog entry documenting the new API product.

The intersection of all three — this one Level × this one Segment × this one Artifact — is a single cell inside the Cube. A real EA repository contains thousands of these cells, one for every level/segment/artifact combination the enterprise has documented.

THE THIRD DIMENSION

Artifacts: The Documentation Itself

Artifacts are the Cube's third dimension — the actual work products that document each Level × Segment intersection. Bernard groups them into three broad types.



Plans & Reports

Written narratives — strategic plans, business cases, and management summaries that explain a Level in prose.



Diagrams & Models

Visual representations — org charts, process flows, data models, and network diagrams, often built with ArchiMate-compatible tools per the syllabus.



Matrices & Inventories

Structured lists and cross-reference tables — application inventories, technology standards lists, and the EA Repository's own six-column grid.

PUTTING IT TOGETHER

How the Three Dimensions Combine

Levels, Segments, and Artifacts aren't three separate lists — they intersect. Every Level, within every Segment, is documented by a set of Artifacts. Change any one coordinate and you're looking at a different part of the enterprise.

CONNECTS TO

- A single Level (e.g. Systems & Applications) repeats once per Segment — Retail Banking's systems are documented separately from Wealth Management's.
- A single Artifact type (e.g. a data model) can exist at multiple Levels — data models appear both at the Data & Information level and inside individual Segments.
- This is why the Cube is drawn in three dimensions rather than as a single flat list — a spreadsheet shows two dimensions at once; the Cube needs three.

COMPARATIVE VIEW

Segments Across Enterprise Types

What counts as a 'Line of Business' changes with the kind of enterprise. Three examples:

Bank (Amanah Bank)

Retail Banking, Commercial Banking, Wealth Management, Digital/FinTech Services — each a Segment with its own five-level documentation.

Government Agency

Segments often follow statutory programs or service lines — e.g. Licensing, Public Health, Revenue Collection — rather than commercial products.

University

Segments can follow academic colleges or delivery modes — e.g. On-Campus Programs, Online-Degree Programs, Continuing Education.

WORKED EXAMPLE

University Online-Degree Initiative: One Cell of the Cube

The course's other running example — the university's online-degree initiative — located inside the Cube the same way Amanah Bank was.



Level

Systems & Applications (Level 4) — the new Learning Management System (LMS) integration.



Segment

Online-Degree Programs — a distinct Line of Business from On-Campus Programs.



Artifact

The Application Inventory entry documenting the LMS integration's interfaces and data flows.

Same structure as the Amanah Bank example, different enterprise: whatever the organization, a cell is always one Level, inside one Segment, documented by one Artifact.

BRIDGE TO WEEK 5

From Cube to Repository: A Crosswalk



The EA Repository (Week 5) isn't a separate structure from the Cube — it's the Cube's Levels and Artifacts organized into a browsable set of columns.

CONNECTS TO

- The Repository's six columns map directly onto the Cube's five Levels plus the Security thread.
- Each Repository column holds the Artifacts for that Level, across every Segment at once.
- Segments become an additional filter on top of the Repository — 'show me only the Retail Banking rows.'

PRACTITIONER NOTES

Common Pitfalls When Using the Cube

Three mistakes that show up often when students — and new architects — first apply the Cube:

Treating Levels as Steps

The five Levels aren't a sequence to complete in order — they're simultaneous views of the same enterprise, updated together.

Ignoring the Threads

Security, Standards, and Skills get forgotten because they don't have their own Level — but skipping them leaves major gaps in the documentation.

One Cube per Company

Large enterprises need one Cube instance per Segment, not one Cube for the whole organization — otherwise Segments blur together.

QUICK RECAP

Six Cube Concepts, Three Contrasts

A last check before we move on:



Vertical vs. Crosscutting Components

Vertical belongs to one Level; crosscutting spans all five (e.g. intranets, buildings & equipment).



Current vs. Future Architecture

Current documents what exists today; Future documents the target state the EA Management Plan is steering toward.



Levels vs. Segments

Levels are horizontal slices — what kind of thing; Segments are vertical slices — which part of the business.

CHAPTER SUMMARY

Key Takeaways

- The EA³ Cube has three dimensions at once: Levels (Week 3), Segments (Lines of Business), and Artifacts (the documentation itself).
- Components can be vertical (specific to one level) or crosscutting (spanning every level — intranets, telecom, buildings).
- The Current → Future Architecture transition, bridged by the Architecture Management Plan, is literally Week 2's six elements seen as one integrated figure.
- Threads — Security, Standards, and Skills — run through every level rather than belonging to any single one.
- Segments connect directly to Week 7's Organizational Network Model: each Line of Business gets its own complete five-level documentation.
- Together, the three dimensions — Levels, Segments, and Artifacts — turn the Cube into the master map of the entire EA Repository. Next week, we look at how AI is beginning to reshape how architects build and maintain that map.

COMING UP NEXT

Further Reading & Next Session

Reading Material

- Bernard, S. (2012). An Introduction to Enterprise Architecture (3rd ed.) — full text, especially the EA³ Cube chapters
- PSU IS321 Syllabus — AI and EA Tools policy
- Weeks 2, 3, 4, 5, 7, 9, 11, and 12 — this week ties directly to all of them

Next: Week 14

AI in Modern Enterprise Architecture — how AI-assisted modeling, AI-supported governance, and smart repositories are layered onto the Cube and TOGAF. Mapped to CLO1, CLO3, CLO5.

Questions?